

NREL 5 MW in laminar sheared inflow

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An NREL 5 MW rotor in a steady, sheared inflow with no turbulence. The flow comes from a PALM large-eddy simulation coupled to OpenFAST (AeroDyn 15), with the rotor represented by an actuator line. This short document explains the OpenFAST output files that come with it.

1. Inflow profile

The inflow is a power-law shear and is the same in every case. The flow is aligned with the streamwise (x) direction only; the lateral and vertical components are zero. The speed grows with height as

$$U(z) = 11.4 \left(\frac{z - 414}{90} \right)^{0.143}$$

with reference speed 11.4 m/s at hub height and shear exponent 0.143. Here z is the height in the PALM domain (m). The hub is at $z = 504$ m, which is 90 m above a virtual ground at $z = 414$ m, so the shear across the rotor matches a real turbine with a 90 m hub height instead of being measured from the domain floor. Below 2 m above the virtual ground the speed is held flat at 6.6145 m/s. The profile is fixed at the inlet and does not change in time. Figure 1 shows it: subfigure (a) over the full domain height, subfigure (b) zoomed on the rotor, where the speed runs from about 9.6 m/s at the bottom of the disk to 11.4 m/s at the hub and 12.3 m/s at the top.

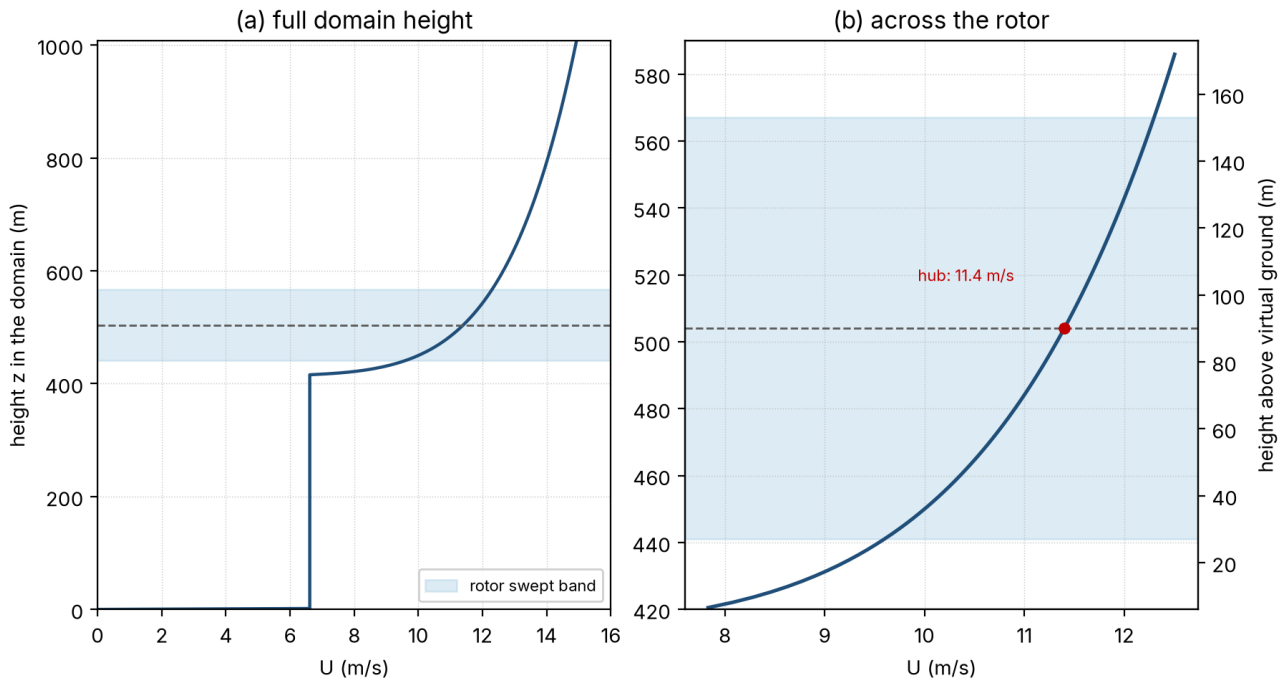


Figure 1. Imposed inflow $U(z)$. The shaded band is the rotor-swept height (441 to 567 m, i.e. hub $\pm R$). The dashed line is hub height (504 m, $U = 11.4$ m/s).

2. Domain and turbine

The domain is a box, with rotor diameter $D = 126$ m and radius $R = 63$ m. The grid is uniform at 1.575 m in all three directions ($R/40$). Table 1 gives the extent, the grid spacing, and where the turbine sits.

Table 1. Domain extent, grid spacing, and turbine position.

Direction	Extent	In diameters	Grid spacing	Turbine position
x, streamwise	1260 m	10 D	$dx = 1.575$ m	rotor 3 D from the inlet
y, lateral	1008 m	8 D	$dy = 1.575$ m	centred (4 D)
z, vertical	1008 m	8 D	$dz = 1.575$ m	hub at 4 D (504 m)

The turbine stands 3 D (378 m) downstream of the inlet, centred across the domain, with the hub at mid-height (4 D = 504 m). That leaves 3 D of clear inflow upstream and 7 D of space for the wake downstream, and about 3.5 D from the rotor edge to each side wall and to the top and bottom. Table 2 lists the rotor geometry and the operating point.

Table 2. Rotor geometry and operating point.

Property	Value
Turbine	NREL 5 MW
Number of blades	3
Rotor diameter, D	126 m
Rotor radius, R (TipRad)	63 m
Hub radius	1.5 m
Hub height	504 m (4 D)
Precone angle	0 deg (all 3 blades, not coned)
Shaft tilt angle	0 deg
Overhang	-5.02 m (rotor upwind of the tower)
Rotor speed	12.1 rpm, fixed (Clockwise rotation)
Blade pitch	0 deg, fixed
Nacelle yaw	0 deg
Structural degrees of freedom	all off (rigid)

All ElastoDyn structural degrees of freedom are off, so the turbine is rigid: the blades and tower do not bend, the deflection channels stay near zero, and the load channels report the reaction forces and moments.

3. Cases

Two things change from case to case. Everything else (inflow, domain, time step, turbine) is the same.

1. Force smearing width (epsilon), used by the actuator-line method to spread the blade forces onto the grid. It is set through $\text{reg_fac} = \text{epsilon} / \text{dx}$ with $\text{dx} = 1.575 \text{ m}$, so reg_fac of 2, 3, 4, 5 gives epsilon of 3.15, 4.725, 6.3, 7.875 m.

2. Near-wake correction, either on or off.

The near-wake correction accounts for the effect of the force smearing itself on the induced velocity; if that is confusing, you can simply ignore the four "nwc" cases and use only the four "no_nwc" ones.

That makes $4 \times 2 = 8$ cases. The case name is in each file name.

Case (name tag)	reg_fac	epsilon (m)	Near-wake correction
nrel5mw_laminar_shear_eps2_nwc	2	3.15	on
nrel5mw_laminar_shear_eps3_nwc	3	4.725	on
nrel5mw_laminar_shear_eps4_nwc	4	6.3	on
nrel5mw_laminar_shear_eps5_nwc	5	7.875	on
nrel5mw_laminar_shear_eps2_no_nwc	2	3.15	off
nrel5mw_laminar_shear_eps3_no_nwc	3	4.725	off
nrel5mw_laminar_shear_eps4_no_nwc	4	6.3	off
nrel5mw_laminar_shear_eps5_no_nwc	5	7.875	off

4. The .out files and their columns

Each *.out* file is plain text. It starts with a few header lines (run information), then two rows: the channel names and their units. After that come the data rows, one per time step, tab separated. The first column is Time (s). Output is written every step (DT = 0.015625 s, that is 1/64 s) and each run covers about 295 s of simulated time.

The remaining columns are OpenFAST output channels, in this order: first the structural channels from ElastoDyn, then the aerodynamic channels from AeroDyn (the rotor totals, then the per blade-node values). The tables below describe the rotor, shaft, blade and aerodynamic channels. The file also has tower-base and yaw-bearing load channels, which are not covered here. A note on sign and coordinate systems: c is the coned blade system, b is the blade system, h is the rotor hub system.

Note. Ignore the initial startup transient. Each run is 295 s long; for steady values, average over the last 20 rotor revolutions.

4.1 Rotor, shaft and nacelle (ElastoDyn)

Channel	Units	Meaning
RotSpeed	rpm	Rotor speed (low-speed shaft). Fixed at 12.1 rpm here.
GenSpeed	rpm	Generator speed (high-speed shaft).
Azimuth	deg	Blade 1 azimuth angle.
BldPitch1	deg	Blade 1 pitch angle. Fixed at 0 here.
YawPos	deg	Nacelle yaw angle. Fixed at 0 here.
RotThrust	kN	Rotor thrust along the shaft axis.
RotTorq	kN-m	Rotor torque on the low-speed shaft. Listed twice in the setup, so it may appear as two identical columns.
RotPwr	kW	Rotor mechanical power.
LSSGagMya, LSSGagMza	kN-m	Low-speed-shaft bending moment at the main bearing, two perpendicular directions.

4.2 Blade 1 loads and deflections (ElastoDyn)

Channel	Units	Meaning
RootFxc1, RootFyc1, RootFzc1	kN	Blade 1 root forces: out-of-plane shear, in-plane shear, axial (coned system).
RootMxc1, RootMyc1, RootMzc1	kN-m	Blade 1 root moments: in-plane bending, out-of-plane bending, pitching (coned system).
RootMxb1, RootMyb1	kN-m	Blade 1 root bending moments in the blade system: edgewise (xb), flapwise (yb).
Spn2MLxb1, Spn2MLyb1	kN-m	Blade 1 local bending moments (edgewise, flapwise) at blade strain-gage station 2, which is ElastoDyn blade node 9 of 62, near $r/R = 0.16$ (low on the blade, not mid-span).
OoPDefl1, IPDefl1	m	Blade 1 tip out-of-plane and in-plane deflection. Near zero (rigid blades).
TwstDefl1	deg	Blade 1 tip twist deflection. Near zero (rigid blades).

4.3 Rotor aerodynamic totals (AeroDyn)

These are AeroDyn's integrated loads on the whole rotor, in the hub system (h). Note the units: AeroDyn reports these in N, N-m and W, while the ElastoDyn rotor loads above are in kN, kN-m and kW.

Channel	Units	Meaning
RtAeroFhx	N	Rotor aerodynamic force along the hub x axis: the aerodynamic thrust.
RtAeroFyh, RtAeroFzh	N	Rotor aerodynamic force, hub y and z axes.
RtAeroMhx	N-m	Rotor aerodynamic moment about the hub x axis: the aerodynamic torque.
RtAeroMyh, RtAeroMzh	N-m	Rotor aerodynamic moment, hub y and z axes.
RtAeroPwr	W	Rotor aerodynamic power.
RtArea	m ²	Rotor swept area (constant).

4.4 Per blade-node aerodynamics (AeroDyn)

These are written for every blade node, on all 3 blades, at all 64 nodes. A channel is named **ABN<nnn><name>**: blade b (1 to 3), node nnn (001 to 064, from root to tip), then the quantity. For example AB2N017Alpha is the angle of attack at blade 2, node 17. That is $3 \times 64 \times 8 = 1536$ columns.

Channel	Units	Meaning
...Alpha	deg	Angle of attack at the section.
...Cl	-	Lift coefficient.
...Cd	-	Drag coefficient.
...Vrel	m/s	Relative flow speed at the section.
...Fx	N/m	Force per unit length normal to the rotor plane (out of plane).
...Fy	N/m	Force per unit length tangential to the rotor plane (in plane).
...DynP	Pa	Dynamic pressure, $0.5 \rho V_{rel}^2$.
...Theta	deg	Section angle: blade pitch plus local twist.

The per-node blade geometry is fixed (the same in every case) and is not in the output file. Table 3 lists it for each node: the radius (r/R and r, with r from the rotor centre and R = 63 m), the chord, the aerodynamic twist, and the airfoil. Node 1 is nearest the root, node 64 the tip.

Table 3. Per-node blade geometry: radius, chord, twist, and airfoil (root to tip).

Node	r/R	r (m)	Chord (m)	Twist (deg)	Airfoil
1	0.024	1.50	3.542	13.308	Cylinder1
2	0.039	2.48	3.542	13.308	Cylinder1
3	0.055	3.45	3.609	13.308	Cylinder1
4	0.070	4.43	3.720	13.308	Cylinder1
5	0.086	5.40	3.832	13.308	Cylinder1
6	0.101	6.38	3.943	13.308	Cylinder1
7	0.117	7.36	4.055	13.308	Cylinder2
8	0.132	8.33	4.167	13.308	Cylinder2
9	0.148	9.31	4.278	13.308	Cylinder2
10	0.163	10.29	4.390	13.308	DU40_A17
11	0.179	11.26	4.501	13.308	DU40_A17
12	0.194	12.24	4.568	13.090	DU40_A17
13	0.210	13.21	4.591	12.655	DU40_A17
14	0.225	14.19	4.614	12.220	DU35_A17
15	0.241	15.17	4.636	11.785	DU35_A17
16	0.256	16.14	4.638	11.386	DU35_A17
17	0.272	17.12	4.592	11.072	DU35_A17
18	0.287	18.10	4.546	10.758	DU35_A17
19	0.303	19.07	4.500	10.444	DU35_A17
20	0.318	20.05	4.453	10.135	DU35_A17
21	0.334	21.02	4.403	9.861	DU35_A17
22	0.349	22.00	4.354	9.587	DU35_A17
23	0.365	22.98	4.304	9.312	DU30_A17
24	0.380	23.95	4.254	9.038	DU30_A17
25	0.396	24.93	4.197	8.750	DU30_A17
26	0.411	25.90	4.140	8.461	DU30_A17
27	0.427	26.88	4.082	8.171	DU25_A17
28	0.442	27.86	4.024	7.882	DU25_A17

Node	r/R	r (m)	Chord (m)	Twist (deg)	Airfoil
29	0.458	28.83	3.964	7.587	DU25_A17
30	0.473	29.81	3.902	7.289	DU25_A17
31	0.489	30.79	3.841	6.991	DU25_A17
32	0.504	31.76	3.779	6.693	DU25_A17
33	0.520	32.74	3.719	6.403	DU25_A17
34	0.535	33.71	3.660	6.122	DU25_A17
35	0.551	34.69	3.602	5.840	DU21_A17
36	0.566	35.67	3.543	5.558	DU21_A17
37	0.582	36.64	3.484	5.277	DU21_A17
38	0.597	37.62	3.426	4.998	DU21_A17
39	0.613	38.60	3.367	4.719	DU21_A17
40	0.628	39.57	3.309	4.439	DU21_A17
41	0.644	40.55	3.250	4.163	DU21_A17
42	0.659	41.52	3.192	3.910	DU21_A17
43	0.675	42.50	3.133	3.657	DU21_A17
44	0.690	43.48	3.074	3.403	NACA64_A17
45	0.706	44.45	3.016	3.150	NACA64_A17
46	0.721	45.43	2.957	2.952	NACA64_A17
47	0.737	46.40	2.899	2.760	NACA64_A17
48	0.752	47.38	2.840	2.568	NACA64_A17
49	0.768	48.36	2.782	2.377	NACA64_A17
50	0.783	49.33	2.723	2.187	NACA64_A17
51	0.799	50.31	2.664	1.998	NACA64_A17
52	0.814	51.29	2.606	1.809	NACA64_A17
53	0.830	52.26	2.547	1.620	NACA64_A17
54	0.845	53.24	2.489	1.431	NACA64_A17
55	0.861	54.21	2.430	1.242	NACA64_A17
56	0.876	55.19	2.372	1.052	NACA64_A17
57	0.892	56.17	2.313	0.863	NACA64_A17
58	0.907	57.14	2.232	0.687	NACA64_A17
59	0.923	58.12	2.151	0.511	NACA64_A17
60	0.938	59.10	2.038	0.351	NACA64_A17
61	0.954	60.07	1.800	0.257	NACA64_A17
62	0.969	61.05	1.562	0.163	NACA64_A17
63	0.985	62.02	1.324	0.001	NACA64_A17
64	1.000	63.00	1.085	-0.260	NACA64_A17